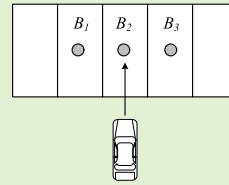


# Collaborative Sensing-Based Parking Tracking System With Wireless Magnetic Sensor Network

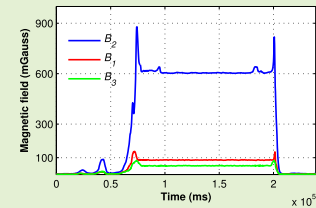
Zusheng Zhang<sup>ID</sup>, Xufei Mao, Kunxiao Zhou, and Huaqiang Yuan

**Abstract**—Wireless magnetic sensor networks provide a low-cost technology for the outdoor real-time parking tracking system. However, the magnetic sensor is susceptible to the interference caused by vehicles parked in adjacent spaces. Most existing detection systems are solely based on local sampled data from the sensor node and they are prone to high false and missed detection rates in the case of a low signal-to-noise ratio. This paper proposes a collaborative sensing parking tracking system by fusing the signals of the adjacent sensors to eliminate the interference. Sensor nodes convert magnetic signals into signatures represented by peak values, and then the base station computes the spatial relationships between adjacent signatures to identify parking event or interference. An experiment system includes sixty sensor nodes that were deployed in bay parking spaces. Experiment results verified that the proposed system has better detection accuracy than existing systems.

**Index Terms**—Magnetic sensor, parking tracking system, collaborative sensing, wireless sensor network.



When a vehicle parked on  $B_2$ , nodes  $B_1$  and  $B_3$  in adjacent spaces also sensed the disturbance of magnetic field.



## I. INTRODUCTION

**P**ROBLEMS arising from the gap between the demand and supply of parking spaces are becoming increasingly acute in most urban environment. An automated parking tracking system is an intelligent transportation system technology where the presence of parked vehicles is monitored in real-time. This technology can be used by drivers to find parking efficiently while minimizing time and effort.

Parking space tracking systems are categorized into two types: indoor and outdoor. Technologies currently employed indoors, such as ultrasonic [1], image (camera) [2], are not suitable for large scale applications in outdoor because they are susceptible to outdoor environmental interference, such as weather conditions, occlusions, and lighting variations. Based on the fact that most vehicles have ferrous metals, a Wireless Magnetic Sensor Network (WMSN) [3]–[5] is a viable and pragmatic alternative for an outdoor parking tracking system.

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WMSN is wireless communication system, battery powered, and its life expectancy is more than 5 years. Recently, wireless magnetic sensor networks are gradually used for outdoor parking detection.

In a WMSN parking tracking system, the sensor node is deployed at the center of each parking space. The sensor samples the geomagnetic field periodically. When a vehicle enters or leaves the space, the sensor detects the status change and generates parking event. However, a vehicle's magnetic field is not only the target of detection but also the source of interference. Adjacent interference is created by vehicles parked on adjacent spaces. There are three types of parking spaces: bay, parallel and angled. In the case of bay parking, the distance between the sensor node and adjacent parked vehicle is shorter than angled and parallel parking. Therefore, adjacent interference is significant and more prominent in bay parking. All of the following discussion of this work is based on the bay parking spaces.

Most of the existing WMSN based parking tracking systems make an occupied or vacant decision solely on the local sampled data of the sensor. The sensors in bay parking spaces are susceptible to the adjacent interference and result in high false and missed detection rates. Zhu *et al.* [6] proposed a hybrid detection algorithm based on local threshold detection and data fusion. However, in the actual parking tracking system, these thresholds of local detection are difficult to accurately estimate. This paper proposes a collaborative sensing based parking tracking system. Our main contributions are given as follows:

(1) The theoretical model of adjacent magnetic signals caused by the parking action of a vehicle is established and the characteristics of the adjacent interference are analyzed.

(2) All parking detections are made by fusing the data of adjacent sensors. Sensor nodes convert magnetic signals into signatures represented by peak values, and BS (Base Station) computes the pattern similarity between adjacent signatures to identify parking event and interference.

(3) We developed a parking tracking system which includes sixty sensor nodes, and experiment results verified that the proposed system has better detection accuracy than existing systems.

The remainder of this paper is organized as follows. Section II surveys the related works. Section III describes the adjacent interference problem and the signal model. Section IV proposes the algorithm for parking detection. Section V conducts experiments to prove the performance of the system. Finally, Section VI makes a brief conclusion.

## II. RELATED WORKS

Parking tracking system with WMSN has been present in both the research community and industry. Threshold detection algorithms [7] usually are used to measure the geomagnetic field change after the vehicle is parked, if the change continuous exceeds a threshold, the sensor will indicate the space is occupied. Cheung *et al.* [8] proposed an adaptive threshold algorithm for parking detection. Ding and Cheung [9] proposed a min-max detection algorithm. Modified min-max algorithm for magnetometers [10] has also been published. Zhu *et al.* [11] proposed a parking detection algorithm by calculating the correlation between the sampled signals and a referential signal. Zhang and Yu [12] proposed an algorithm for parking occupancy detection based on the distance discriminate analysis method.

Some works have studied the problem of adjacent interference. Trigona *et al.* [13] conducted the magnetic interference test of the adjacent vehicle in the bay parking spaces. Saifil *et al.* [14] found that the Z-axis was less affected by interference. Liu and Zheng [15] analyzed the interference problem caused by the neighboring vehicles and studied the influence of parking direction and sensor coordinate direction on the magnetic field strength. These methods choose the signal with the least interference (e.g. Z-axis' signal) for vehicle detection. They ignore the spatial correlation of adjacent signals and are not reliable.

Zhu *et al.* [6] proposed a parking detection method includes two parts: 1) Local detection algorithm, each node uses hard thresholds and state machine mechanism for parking detection; 2) Data fusion based on Dempster-Shafer evidence, when the local detection algorithm cannot achieve accurate detection, they use the Dempster-Shafer evidence theory for the final detection. However, their algorithm is mainly based on local data detection, and more than 90% of the parking detection only depends on local data for decision-making. The hard thresholds for local detection are related to many factors, such as the actual deployment environment, the model of the sensor, the gain settings of sensor (e.g., there are eight settings for

the gain of HMC5983), and so on. The disadvantage of this method is that its hard thresholds are difficult to estimate accurately.

Some researchers propose to add other types of sensors for auxiliary detection. Farhad *et al.* [16] employed four sensors, magnetic, light, temperature, and ambient temperature sensors of the motes for reliable detection of the free space. Sifuentes *et al.* [17] designed a node based on two sensors (one magnetic and one optical) to detect the presence of a vehicle. When a vehicle arrives or departs, an optical sensor detects the change of light and wakes up the magnetic sensor which measures the magnetic field to detect the vehicle. However, these sensors are often susceptible to the outdoor physical interference such as a dust blocked optical sensor which can't work effectively.

## III. PROBLEM DEFINITION

Considering a parking tracking WMSN  $G = (B, R, BS)$ , where  $B = \{B_1, B_2, \dots, B_n\}$  is the set of sensor nodes,  $R = \{R_1, R_2, \dots, R_m\}$  is the set of routers, and  $BS$  denotes a base station. The discrete sampled signal  $M(t)$  from a sensor node  $B_i$  can be expressed as equation 1.

$$M_\gamma(t) = E_\gamma(t) + V_\gamma(t) + I_\gamma(t) \quad (1)$$

where  $t$  denotes the sample time,  $\gamma$  indicates the index of sensor's axis,  $E_\gamma(t)$  is the geomagnetic field which is uniform over a wide area on a scale of kilometers,  $V_\gamma(t)$  is the target magnetic signal created by a parking vehicle,  $I_\gamma(t)$  is the interference signal.

### A. Adjacent Interference Problem

The sensor used is a three-axis AMR (Anisotropic Magneto-Resistive) sensor QMC5883 [18]. In order to reduce the dimension of the raw signal, the three-axis signals are synthesized as a fused signal using equation (2).

$$S(t) = \sum_{\gamma} |M_\gamma(t) - E_\gamma(t)| \quad \gamma \in (x, y, z) \quad (2)$$

The earth's magnetic field has been drifting westward, and the change is minimal for several months [19]. Therefore, in our system, BS issues an instruction to update  $E_\gamma$  every two months.  $E_\gamma$  is approximately equal to the average value of samples when the status of  $B_i$  and its adjacent parking spaces are all *vacant*.

As shown in Fig. 1,  $B_1 = (0, w, 0)$  and  $B_2 = (0, 0, 0)$  represent two adjacent nodes deployed in the parking spaces, where  $w$  is the width of a parking space. Vehicles with high magnetic flux density can interfere with sensors in adjacent parking spaces. For example, when a Cadillac SUV had parked in  $B_2$ , Fig. 2 shows the signals  $S_1$  and  $S_2$  from nodes  $B_1$  and  $B_2$ , respectively. The signal  $S'_1$  shown in Fig. 3 is a signal of a mini vehicle parked in  $B_1$ . Compare signals  $S_1$  (in Fig. 2) and  $S'_1$  (in Fig. 3), if only based on the local data of the sensor node, it can't make a correct decision whether there is a target signal or adjacent interference. Therefore, it is necessary to fuse the adjacent signals to identify the interference.

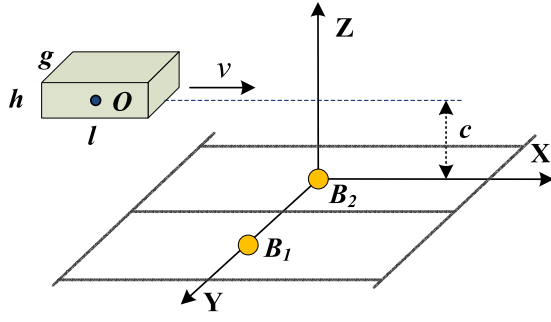


Fig. 1. Model of a vehicle parking.

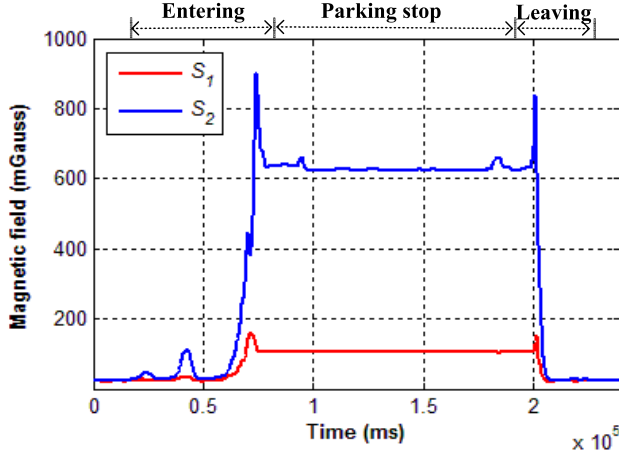


Fig. 2. Experiment signals of adjacent nodes.

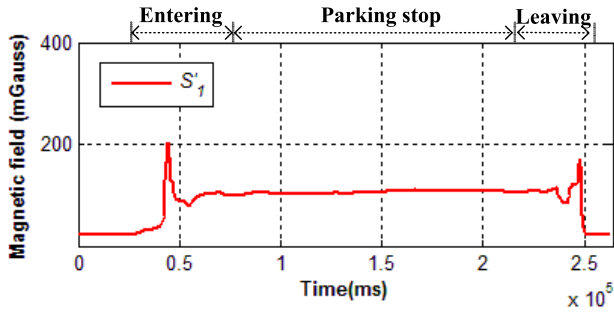


Fig. 3. The magnetic signal of a smart car.

As shown in Fig. 2, a whole parking period includes three phases: entering, parking and leaving. In the entering or leaving phases, the signal fluctuates continuously. When a node  $B_i$  detects the fluctuation caused by a vehicle enters or leaves,  $B_i$  extracts the peak values (including the initial value, local maxima, local minima, and the terminal value) as the signal's signature [12], [20] and sends it to BS. BS uses a Collaborative Sensing Detection Algorithm (CSDA) to decide the parking space status. Presuming the ID of a parking space is same with the correspond node's ID, the algorithm can be defined as following:

**Definition 1: Collaborative sensing**

$R = \text{CSDA}(U_i, U_{i-1}, U_{i+1})$ .

**Input:**  $U_i$  is  $B_i$ 's signature,  $U_{i-1}$  and  $U_{i+1}$  are the signature vectors from the adjacent nodes  $B_{i-1}$  and  $B_{i+1}$ .

**Task:** The algorithm needs to accurately identify the adjacent interference signal and the target signal in order to realize the accurate detection of the parking status.

**Output:**  $R = 1$ , the parking space  $B_i$  is {occupied};  $R = 0$ , the parking space  $B_i$  is {vacant}.

### B. Signal Model of Adjacent Interference

Generally, the magnetic field of a vehicle can be approximated by the field created by a rectangular magnet. Fig. 1 shows a rectangle magnet with length  $l$ , width  $g$ , height  $h$  and a uniform magnetization in its body under the saturation state. The point  $O = (a, b, c)$  is the initial position of a vehicle. According to the magnetic field model of rectangle magnet [21], the magnetic signals created by a vehicle can be calculated as equation 3.

$$S(t) = \frac{K}{2} [\Gamma(x(t), y(t), z(t)) - \Gamma(\frac{l}{2} - x(t), y(t), z(t)) - \Gamma(x(t), g - y(t), z(t)) + \Gamma(\frac{l}{2} - x(t), g - y(t), z(t))] \Big|_0^h \quad (3)$$

where  $K = \mu_0 J / 4\pi$ ,  $\mu_0$  is the air magnetic permeability,  $J$  is the surface current density,  $x(t)$ ,  $y(t)$ ,  $z(t)$  are the distance functions between vehicle coordinates and node coordinates on the three axes of  $x$ ,  $y$ ,  $z$ , respectively.

$$\Gamma(r_1, r_2, r_3) = \ln \frac{\sqrt{r_1^2 + r_2^2 + (r_3 - \hat{z})^2} - r_2}{\sqrt{r_1^2 + r_2^2 + (r_3 - \hat{z})^2} + r_2} \quad (4)$$

where  $[\bullet]_0^h$  represents the difference of function in the bracket when  $\hat{z} = h$  and  $\hat{z} = 0$ .

Given the linear motion of the vehicle along the X-axis, the parking process is as follows: at time  $t = 0$ , a vehicle starts from  $O$  and enters the  $B_2$  with velocity  $v_0$ , then it parks on the sensor  $B_2$  for a period of  $T$ , and then it leaves  $B_2$ . Assuming the vehicle's acceleration is uniform during the entering and leaving phases, i.e.  $c$  is acceleration and  $c = v_0^2 / 2a$ . For node  $B_1$ , the functions can be expressed as  $\{x_1(t), y_1(t) = w - b, z_1(t) = c\}$ , where  $x_1(t)$  is shown in equation (5). The functions for node  $B_2$  are  $\{x_2(t) = x_1(t), y_2(t) = b, z_2(t) = c\}$ .

$$x_1(t) = \begin{cases} a + \int_0^t (v_0 - ct) dt & \text{if } 0 < t \leq T_e \\ 0 & \text{if } T_e < t \leq T_s \\ -\int_{T_s}^t (c * (t - T_s)) dt & \text{if } T_s < t \leq T_l \end{cases} \quad (5)$$

where  $T_e = v_0 / c$ ,  $T_s = T_e + T$  and  $T_l = T_s + T_e$  are the end times of a vehicle's entering, parking and leaving phases, respectively.

Assume  $O = (-8, 0.2, 0.2)$ , the speed is  $v_0 = 5m/s$ , the parking stop time is  $T = 5s$ , the sampling frequency is  $f = 200Hz$ , and the measurement noise is Gaussian white noise. Fig. 4 shows the simulated signals by using the signal model when a vehicle parked in  $B_2$ . The simulated patterns are in agreement with the experiment measurements. We found that the adjacent signals caused by a vehicle parking have the following characteristics.

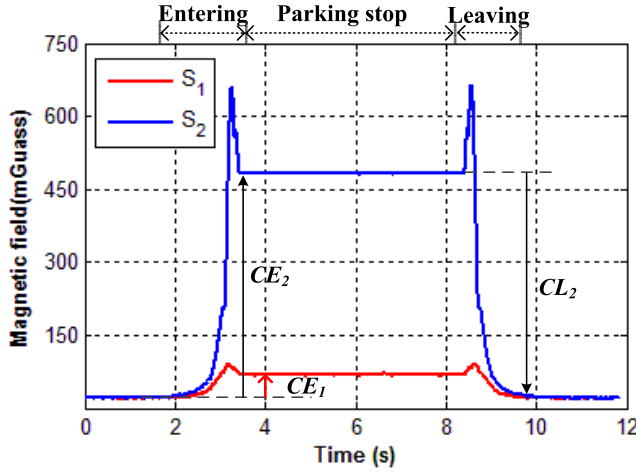


Fig. 4. Simulation signals of adjacent nodes.

**Definition 2: Characteristics of adjacent signals:** as shown in Fig. 4,  $S_1$  and  $S_2$  are the adjacent signals,  $S_1$  is the interference signal,  $S_2$  is the target signal.

(1) *Time synchronization:* during the entering and leaving phases, the fluctuation of  $S_1$  and  $S_2$  are basically synchronized in the time domain.

(2) *Change ratio:* the function  $Change()$  records the difference between the last and the first value of a continuous fluctuation. In the entering phase, the  $Change(Entering(S_1))$  is much less than  $Change(Entering(S_2))$ , i.e.  $CE_1 \ll CE_2$ . It is same in the leaving phase.

(3) *Pattern similarity:* the waveform of  $S_1$  and  $S_2$  are similar during the entering and leaving phases.

#### IV. ALGORITHM

In this section, we first introduce the similarity calculation method, then describe the collaborative sensing algorithm.

##### A. Similarity Calculation

Given  $U_1$  and  $U_2$  are the received signatures from two adjacent nodes, BS calculates their similarity according to the definition 2.

$$U_1 = \{(i, t_i, x_i), 0 \leq i \leq N\} \quad (6)$$

$$U_2 = \{(j, h_j, y_j), 0 \leq j \leq M\} \quad (7)$$

where  $i$  and  $j$  are the counters,  $t_i$  and  $h_j$  are the sample times,  $x_i$  and  $y_j$  are the peak values.

1) *Time Synchronization:* If the time of  $U_1$  and  $U_2$  are basically synchronized, there has  $Syn(U_1, U_2) = 1$ , otherwise,  $Syn(U_1, U_2) = 0$ .

$$Syn(U_1, U_2) = \begin{cases} 1 & \text{if } |t_0 - h_0| \approx 0 \text{ \& \& } |t_n - h_m| \approx 0 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

where  $t_0$  and  $t_n$  are the start time and end time of  $U_1$ ,  $h_0$  and  $h_m$  are the start time and end time of  $U_2$ .

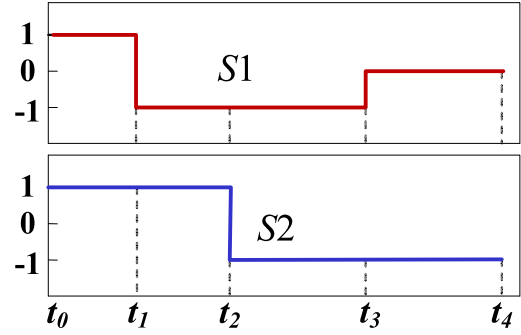


Fig. 5. Two pattern sequences.

2) *Change Ratio:* If the  $Change(U_2)$  is far greater than  $Change(U_1)$ , there has  $Ratio(U_1, U_2) = 1$ . Otherwise,  $Ratio(U_1, U_2) = 0$ .

$$Ratio(U_1, U_2) = \begin{cases} 1 & \text{if } Change(U_2)/Change(U_1) \geq G_1 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where  $G_1$  is a given constant,  $Change(U_2) = (y_m - y_0)$  and  $Change(U_1) = (x_n - x_0)$ .

3) *Pattern Similarity:* Pattern distance is often used to calculate the morphological similarity of time series [22]. A pattern represents a tendency. The pattern is a triple  $\{Up, Hold, Down\}$ , and can also be expressed as  $P = \{1, 0, -1\}$ . The signature is transformed into a pattern sequence according to the *Slope* values, following use  $U_1$  as an example.

$$Slope_i = \frac{x_i - x_{i-1}}{t_i - t_{i-1}} \quad (10)$$

$$p_i = \begin{cases} 1 & Slope_i \geq +g * G_2 \\ -1 & Slope_i \leq -g * G_2 \\ 0 & \text{Otherwise} \end{cases} \quad (11)$$

where  $G_2$  is a constant and  $g = \sqrt{Change(U)}$ . The pattern sequence is expressed as an order set of the (*pattern, time*) pair:

$$U = \{(\wedge, t_0), (p_1, t_1), \dots, (p_i, t_i), \dots, (p_n, t_n)\} \quad (12)$$

where  $p_i \in P$ ,  $\wedge$  indicates *null*,  $t_i$  ( $i = 1, 2, \dots, n$ ) is the pattern ending time,  $n$  is the number of pattern segments. Pattern  $p_i$  keeps from  $t_{i-1}$  to  $t_i$ .

Usually, the time segments between two pattern sequence from adjacent nodes are not consistent. Fig. 5 shows the pattern representation of two sequences  $S1$  and  $S2$ . Their time segments are inconsistent. Therefore, it is necessary to perform a time equivalent process before calculating the pattern distance.

$$S1 = \{(\wedge, t_0), (1, t_1), (-1, t_3), (0, t_4)\} \quad (13)$$

$$S2 = \{(\wedge, t_0), (1, t_2), (-1, t_4)\} \quad (14)$$

The time equivalent process is to re-divide the line interval according to the union of the extremes of two series.



After the time equivalent process, the results of  $S1$  and  $S2$  are as equations (15) and (16):

$$S1 = \{(\wedge, t_0), (1, t_1), (-1, t_2), (-1, t_3), (0, t_4)\} \quad (15)$$

$$S2 = \{(\wedge, t_0), (1, t_1), (1, t_2), (-1, t_3), (-1, t_4)\} \quad (16)$$

Presume the transformed  $U_1$  and  $U_2$  are pattern sequences whose segments are the same in the time domain. The pattern distance can be computed as equation (19).

$$U_1 = \{(\wedge, t_0), (p'_1, t_1), \dots, (p'_i, t_i), \dots, (p'_n, t_n)\} \quad (17)$$

$$U_2 = \{(\wedge, t_0), (p''_1, t_1), \dots, (p''_i, t_i), \dots, (p''_n, t_n)\} \quad (18)$$

$$D(U_1, U_2) = \frac{1}{T_s} \left( \sum_{i=1}^n (t_i - t_{i-1}) * |p'_i - p''_i| \right) \quad (19)$$

$$T_s = \sum_{i=1}^n (t_i - t_{i-1}) \quad (20)$$

where  $T_s$  denotes the length of time series,  $t_{ih} = (t_i - t_{i-1})$  denotes the hold time of the  $i$ -th pattern,  $n$  is the total numbers of patterns.  $t_{ih}$  is a weight, the longer the hold time, the more ratios it has in the distance.

The pattern similarity can be describes as: if the waveforms of  $U_1$  and  $U_2$  are basically similar, there has  $Sim(U_1, U_2) = 1$ . Otherwise,  $Sim(U_1, U_2) = 0$ .

$$Sim(U_1, U_2) = \begin{cases} 1 & \text{if } D(U_1, U_2) < G_3 \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

where  $G_3$  is the given constant,  $D(U_1, U_2)$  is the pattern distance.

So it is determined that  $U_1$  is an adjacent interference if  $Similarity(U_1, U_2) = 1$ . Otherwise, it is a target signal.

$$Similarity(U_1, U_2) = \begin{cases} 1 & \text{if } Syn(U_1, U_2) = 1 \\ & \text{and } Ratio(U_1, U_2) = 1 \\ & \text{and } Sim(U_1, U_2) = 1 \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

## B. Collaborative Sensing Algorithm

This subsection introduces the CSDA (Collaborative Sensing Detection Algorithm) according to the definition 1. If the signal fluctuates, node  $B_i$  extracts the signature and sends the vector  $U_i$  to BS. When BS receives the signature from  $B_i$  at time  $T_r$ , it acquires signatures from adjacent nodes of  $B_i$  in time period  $(T_r - t_w) \sim (T_r + t_w)$ , where  $t_w$  is the time of a waiting timer. When the signatures of adjacent nodes are received or the waiting timer is timed out, BS performs the collaborative sensing calculation.

Assuming node  $B_i$ 's neighbors are  $B_{i-1}$  and  $B_{i+1}$ .  $U_i$ ,  $U_{i-1}$  and  $U_{i+1}$  are the signature sequences of  $B_i$ ,  $B_{i-1}$  and  $B_{i+1}$ , respectively. If  $Similarity(U_i, U_{i-1}) = 0 \& \& Similarity(U_i, U_{i+1}) = 0$ ,  $U_i$  is a target magnetic signal created by a parking vehicle, i.e. it isn't a adjacent interference signal which is caused by a vehicle parking on  $B_{i-1}$  or  $B_{i+1}$ . The pseudo code of the collaborative sensing is shown in algorithm 1.

## Algorithm 1 The Collaborative Sensing Algorithm

```

1 Function  $R = CSDA(U_i, U_{i-1}, U_{i+1})$ ;
   Input :  $U_i$  is the received signature of node  $B_i$ ;  $U_{i-1}$ 
           and  $U_{i+1}$  are the signatures for  $B_i$ 's adjacent
           nodes.
   Output:  $R = 0$ : the space  $B_i$  is vacant;
            $R = 1$ : the space  $B_i$  is occupied;
   /*  $ST$  is the latest status of  $B_i$  */
   /*  $CE$  is the entering change of  $B_i$  */
2  $C = Change(U_i)$ ;
3 if  $Similarity(U_i, U_{i-1}) = 0 \& \& Similarity(U_i, U_{i+1}) =$ 
   $0 \& \& |C| > 20$  then
4   if  $ST = 0$  then
5      $CE = C$ 
6      $R = 1$ ;
7   else
8     if  $(CE + C) \approx 0$  then
9        $CE = 0$ ;
10       $R = 0$ ;
11    end
12  end
13 else
14    $R = ST$ ;
15 end
16 return  $R$ ;

```

Parking space's status detection includes two events:  $\{occupied\}$  and  $\{vacant\}$ . Given  $ST$  is the latest status of  $B_i$ , the criteria for  $\{occupied\}$  event detection can be described as follows: if  $ST = 0$ ,  $U_i$  is not an interference signal and  $Change(U_i) > 20$ , the status is  $\{occupied\}$ ,  $R = 1$ .

As shown in Fig. 4, when the vehicle left, there has  $Change(Entering(S_2)) + Change(Leaving(S_2)) \approx 0$ , i.e.  $CE_2 + CL_2 \approx 0$ . Given  $CE$  is the entering change of  $B_i$ , the criteria for  $\{vacant\}$  detection can be described as follows: if  $ST = 1$ ,  $U_i$  is not an interference signal and  $(CE + Change(U_i)) \approx 0$ , the status is  $\{vacant\}$ ,  $R = 0$ .

## V. EXPERIMENTS

### A. Experiment Scenario and Results

To test the proposed system, we deployed sixty sensor nodes in outdoor bay parking spaces. As shown in Fig. 6, the node is using the STM8L151C8 MCU, LoRa module SX1278 communication module, QMC5883 magnetic sensor [18], and the Contiki operating system [23]. As shown in Fig. 7, we used high strength PVC as node shell for resisting the crush of the parking vehicles. Nodes are placed at the center of each parking space.

We developed a parking tracking system as shown in Fig. 8. The node colored as a green dot indicates the correspond parking space is vacant; a red dot means the parking space is occupied; the lines with an arrow describe the wireless network topology. Parameters used in our experiments are given in table I. The values of the parameters  $G_1$ ,  $G_2$ ,  $G_3$  in table I are trained by the collected sample magnetic field signals.



Fig. 6. A magnetic sensor node.



Fig. 7. The shell of a sensor node.



Fig. 8. A parking tracking system.

TABLE I  
EXPERIMENT PARAMETERS

| Parameter | Description                  | Value |
|-----------|------------------------------|-------|
| $G_1$     | Ratio of <i>Change</i> value | 5     |
| $G_2$     | A threshold of slope value   | 0.5   |
| $G_3$     | Pattern distance threshold   | 30%   |
| $t_w$     | a waiting timer              | 1.5s  |

The value of waiting timer  $t_w$  is an empirical value, which is related to the communication delay of the network, and the communication delay is usually no more than 1 second.

Fig. 9 shows the parking detection experiment scene. In our experiments, using manual recording and video monitoring, we can obtain the exact status of the parking spaces. The performance of the CSDA is analyzed using the accuracy of detection.

The experiment system has operated for more than two years. As shown in table II, there have three cases: no adjacent interference, interference from one adjacent space, and interference from two adjacent spaces, the detection accuracy of these cases are 99.85%, 99.57%, and 99.25%, respectively.



Fig. 9. Experiment scene.

TABLE II  
DETECTION ACCURACY

| Case | Situation                             | Number of test | Accuracy |
|------|---------------------------------------|----------------|----------|
| 1    | No adjacent interference              | 2200           | 99.85%   |
| 2    | Interference from one adjacent space  | 3000           | 99.57%   |
| 3    | Interference from two adjacent spaces | 2500           | 99.25%   |

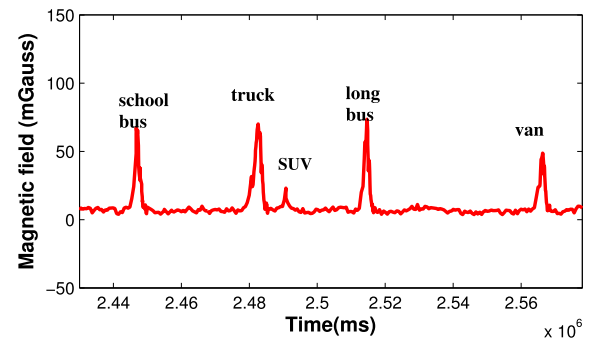


Fig. 10. The interference signals produced by moving vehicles.

### B. Interference of Moving Vehicle

The moving vehicle in the adjacent road produces the magnetic interference signal, and the interference can be discuss as following two cases:

(1) While the status of a sensor node is 'vacant', some vehicles which have high-intensity magnetic field passed in the adjacent lane. As shown in Fig. 10, the magnetic interference signals, generated by moving vehicles, are pulse signals. The characteristic of a stable change is a necessary condition for 'occupied' event detection, so that pulse signals produced by moving vehicles will not affect the 'occupied' event detection.

(2) While the status of a sensor node is 'occupied', there has interference caused by moving vehicle in the adjacent road. As shown in Fig. 11, the interference signal is a pulse signal. The criteria for 'vacant' vent detection is as follow: if a vehicle left, there must has  $Change(Entering) + Change(Leaving) \approx 0$ , take Fig. 11 for example,  $CE + CL \approx 0$ . So pulse signals produced by moving vehicles do not lead to false detection of 'vacant' event.

### C. Performance Comparison

In order to facilitate the performance analysis of the algorithm, the original sampling data of the magnetic field

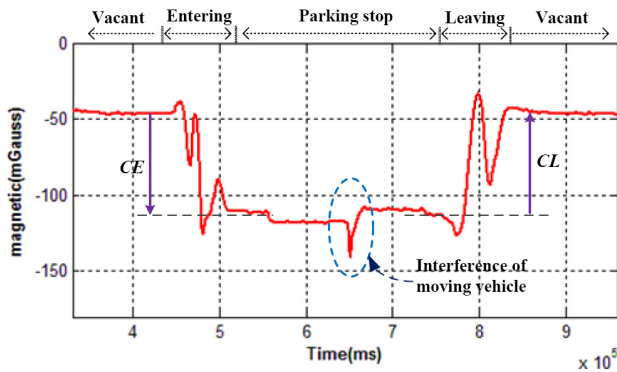
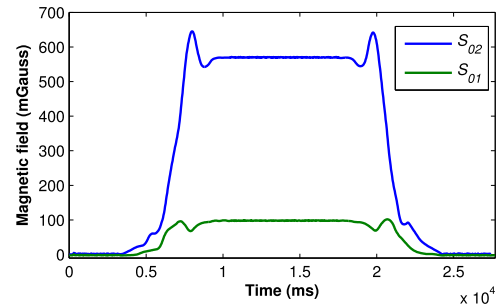


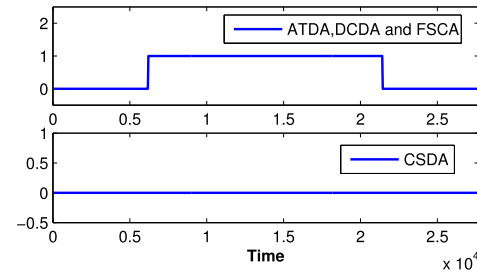
Fig. 11. The magnetic signal of vehicle parking and a interference signal of a moving vehicle.

TABLE III  
PERFORMANCE COMPARISON OF ALGORITHMS

| Algorithm | Number of parking events | Accuracy |
|-----------|--------------------------|----------|
| ATDA      | 5000                     | 89.56%   |
| DCDA      | 5000                     | 93.06%   |
| FSCA      | 5000                     | 97.56%   |
| CSDA      | 5000                     | 99.54%   |

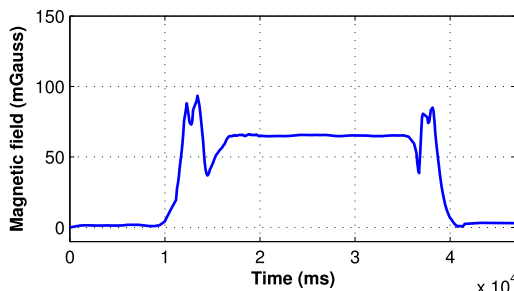


(a) Adjacent signals from nodes  $B_{01}$  and  $B_{02}$

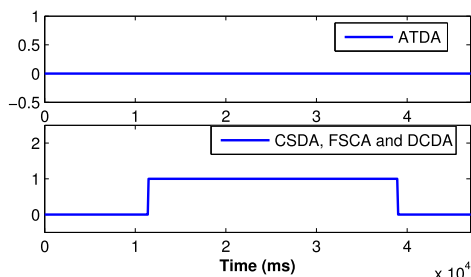


(b) Detection result of signal  $S_{01}$

Fig. 13. The first case of adjacent interference.



(a) Signal from nodes  $B_{15}$



(b) Detection result of the signal

Fig. 12. The weak signal of a vehicle parking.

is packaged and uploaded. Based on the original magnetic data, algorithms ATDA (Adaptive Threshold Detection Algorithm) [8], DCDA (Distance-based Classification Detection Algorithm) [12] and FSCA (Finite-State and Collaborative Algorithm) [6] are adopted for performance comparison with the proposed CSDA. ATDA is a popular parking detection algorithm which is based on the threshold and state machine mechanisms. In order to reduce the missed detections of the weak magnetic signal vehicles, the threshold of ATDA is

equal to 75. DCDA is a distance-based classification detection algorithm that we proposed in 2015, and it has good detection accuracy in parallel parking spaces. According to the refer [6], the thresholds of FSCA's local detection are  $T_{max} = 80$  and  $T_{min} = 40$ .

We performed performance comparison experiments of three algorithms. For each algorithm, the input data included 5000 parking events. As shown in table III, the average detection accuracy of ATDA, DCDA, FSCA and CSDA are 89.56%, 93.06%, 97.56% and 99.54%, respectively. Magnetic signals in bay parking are susceptible to adjacent interference. ATDA and DCDA only use the local data of the sensor node, in the case of low signal to noise ratio, they can't make a correct decision whether there is a target signal or adjacent interference. The selection of hard thresholds has a great influence on the performance of FSCA.

As shown in Fig. 12(a), the magnetic signal from node B15 fluctuates, but these signals from neighbor nodes B14 and B16 do not fluctuate. So it is a parking signal without adjacent interference. Fig. 12(b) displays the detection results of algorithms as proposed above, where 0 and 1 indicate the parking status vacant and occupied, respectively. The ATDA results in missed detections when the target signal less than its threshold.

Fig. 13(a) shows the sampled signals  $S_{01}$  and  $S_{02}$  of two adjacent nodes  $B_{01}$  and  $B_{02}$ , respectively. The signal of  $S_{01}$  is an adjacent interference. The detection results of  $S_{01}$  are shown in Fig. 13(b). The ATDA and FSCA result in false detections when the adjacent interference signal higher than its threshold. The amplitude of fluctuation in the entering phase is bigger than DCDA's threshold, so the detection result of DCDA is '1'. CSDA fuses the data of the adjacent sensors to identify the adjacent interference, and its result is correct.

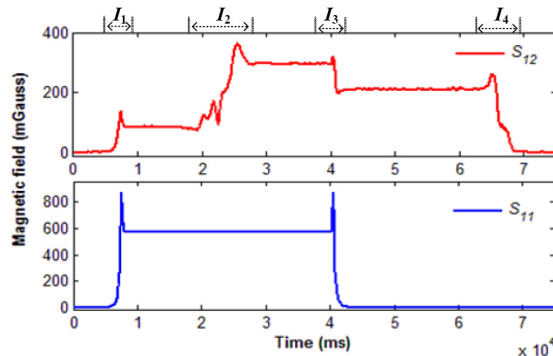
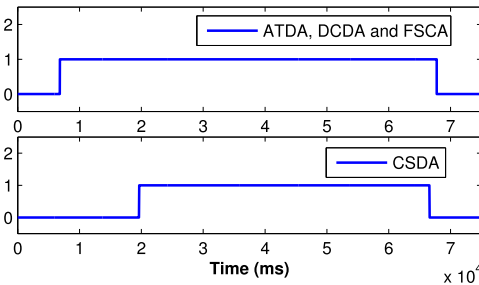
(a) Signals of  $S_{11}$  and  $S_{12}$ (b) Detection results of  $S_{12}$ 

Fig. 14. The second case of adjacent interference.

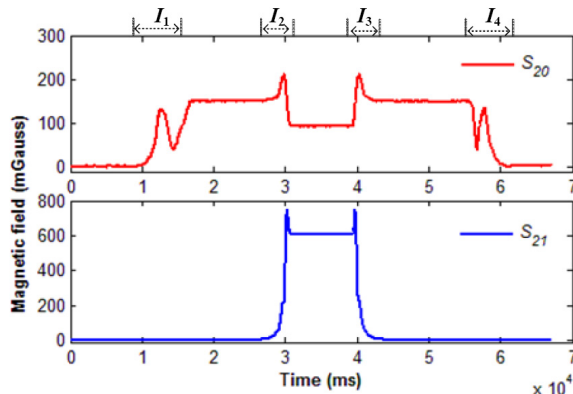
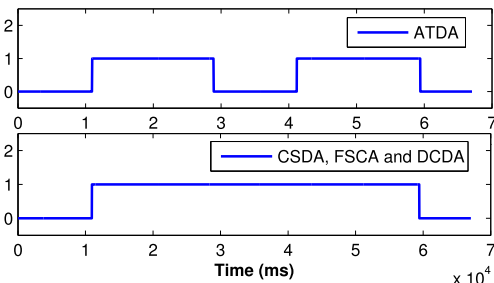
(a) Signals of  $S_{20}$  and  $S_{21}$ (b) Detection result of  $S_{20}$ 

Fig. 15. The third case of adjacent interference.

Fig. 14(a) shows the sampled signals  $S_{11}$  and  $S_{12}$  of adjacent nodes  $B_{11}$  and  $B_{12}$ , respectively. The sequence number of the pulse is marked above the signal graph. For the signal  $S_{12}$ , pulses  $I_1$  and  $I_3$  are the adjacent interferences caused by a

vehicle parked in the  $B_{11}$  parking space. Pulses  $I_2$  and  $I_4$  are the entering and leaving signals caused by a vehicle parked in the  $B_{12}$  parking space, so  $I_2$  and  $I_4$  are the detection targets. The detection results of  $S_{12}$  is shown in Fig. 14(b). Algorithms ATDA, DCDA and FSCA generate false detections of the pulse  $I_1$ , and only CSDA produces the correct results.

Fig. 15(a) shows another case of adjacent interference. For the signal  $S_{20}$ ,  $I_1$  and  $I_4$  are the pulses caused by the entering and leaving of the vehicle parked in the  $B_{20}$  parking space,  $I_2$  and  $I_3$  are interferences caused by a vehicle parked in the  $B_{21}$  parking space, so  $I_1$  and  $I_4$  are the detection targets. The detection results are shown in Fig. 15(b): ATDA generates missed detections because the magnetic field signal is below the threshold between  $I_2$  and  $I_3$ .

## VI. CONCLUSION

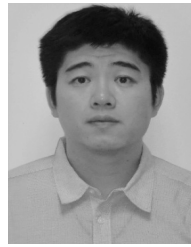
Recently, WMSN is being gradually used as an outdoor parking tracking system. However, the magnetic signal is susceptible to the interference by vehicles parked in adjacent spaces. This paper proposes a collaborative sensing based tracking system to eliminate adjacent interference. Sensor nodes convert magnetic signals into signatures represented by peak values, and BS computes the pattern similarity between adjacent signatures to identify parking event and interference. Experiment results show that the system has better detection accuracy than existing systems.

## REFERENCES

- [1] A. Sahfutri *et al.*, "Smart parking using wireless sensor network system," in *Proc. Int. Conf. Electr. Eng. Comput. Sci. (ICECOS)*, Oct. 2018, pp. 117–122.
- [2] S.-E. Shih and W.-H. Tsai, "A convenient vision-based system for automatic detection of parking spaces in indoor parking lots using wide-angle cameras," *IEEE Trans. Veh. Technol.*, vol. 63, no. 6, pp. 2521–2532, Jul. 2014.
- [3] Q. Wang, J. Zheng, H. Xu, B. Xu, and R. Chen, "Roadside magnetic sensor system for vehicle detection in urban environments," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 5, pp. 1365–1374, May 2018.
- [4] J. Li, S. Cheng, Z. Cai, J. Yu, C. Wang, and Y. Li, "Approximate holistic aggregation in wireless sensor networks," *ACM Trans. Sensor Netw.*, vol. 13, no. 2, p. 11, Aug. 2017.
- [5] Z. He, Z. Cai, S. Cheng, and X. Wang, "Approximate aggregation for tracking quantiles and range countings in wireless sensor networks," *Theor. Comput. Sci.*, vol. 607, pp. 381–390, Nov. 2015.
- [6] H. Zhu, S. Feng, and F. Yu, "Parking detection method based on finite-state machine and collaborative decision-making," *IEEE Sensors J.*, vol. 18, no. 23, pp. 9829–9839, Dec. 2018.
- [7] A. Haoui, R. Kavalier, and P. Varaiya, "Wireless magnetic sensors for traffic surveillance," *Transp. Res. C, Emerg. technol.*, vol. 16, no. 3, pp. 294–306, Jun. 2008.
- [8] S. Cheung and P. Varaiya, "Traffic surveillance by wireless sensor networks: Final report," California PATH, Univ. California, Berkeley, CA, USA, Tech. Rep. UCB-ITS-PRR-2007-4, 2007.
- [9] J. Ding, S.-Y. Cheung, C.-W. Tan, and P. Varaiya, "Signal processing of sensor node data for vehicle detection," in *Proc. 7th Int. IEEE Conf. Intell. Transp. Syst.*, Apr. 2005, pp. 70–75.
- [10] J. Gu, Z. Zhang, F. Yu, and Q. Liu, "Design and implementation of a street parking system using wireless sensor networks," in *Proc. IEEE 10th Int. Conf. Ind. Informat.*, Jul. 2012, pp. 1212–1217.
- [11] H. Zhu and F. Yu, "A vehicle parking detection method based on correlation of magnetic signals," *Int. J. Distrib. Sensor Netw.*, vol. 11, no. 7, Jul. 2015, Art. no. 361242.
- [12] Z. Zhang, M. Tao, and H. Yuan, "A parking occupancy detection algorithm based on AMR sensor," *IEEE Sensors J.*, vol. 15, no. 2, pp. 1261–1269, Feb. 2015.



- [13] C. Trigona *et al.*, "Implementation and characterization of a smart parking system based on 3-axis magnetic sensors," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf.*, May 2016, pp. 1–6.
- [14] S. A. A'ssri, F. H. K. Zaman, and S. Mubdi, "The efficient parking bay allocation and management system using LoRaWAN," in *Proc. IEEE 8th Control Syst. Graduate Res. Colloq. (ICSGRC)*, Aug. 2017, pp. 127–131.
- [15] Y. Liu and L. Zheng, "Research of anti-interference technology based on magnetic field variation of vehicle," in *Proc. 8th Int. Symp. Comput. Intell. Design (ISCID)*, Dec. 2015, pp. 397–400.
- [16] S. Farhad, I. Alahi, and M. M. Islam, "Internet of Things based free parking space management system," in *Proc. Int. Conf. Cloud Comput. Res. Innov. (ICCCRI)*, Apr. 2017, pp. 1–6.
- [17] E. Sifuentes, O. Casas, and R. Pallas-Areny, "Wireless magnetic sensor node for vehicle detection with optical wake-up," *IEEE Sensors J.*, vol. 11, no. 8, pp. 1669–1676, Aug. 2011.
- [18] (2020). *Datasheet of qmc5983*. Qst corp group. [Online]. Available: <http://www.qstcorp.com/productinfo/153108.html>
- [19] M. Dumberry and C. C. Finlay, "Eastward and westward drift of the Earth's magnetic field for the last three millennia," *Earth Planetary Sci. Lett.*, vol. 254, nos. 1–2, pp. 146–157, Feb. 2007.
- [20] K. Kwong, R. Kavalier, R. Rajagopal, and P. Varaiya, "Real-time measurement of link vehicle count and travel time in a road network," *IEEE Trans. Intell. Transp. Syst.*, vol. 11, no. 4, pp. 814–825, Dec. 2010.
- [21] W. Yang, C. Hu, M. Q. H. Meng, H. Dai, and D. Chen, "A new 6D magnetic localization technique for wireless capsule endoscope based on a rectangle magnet," *Chin. J. Electron.*, vol. 19, no. 2, pp. 360–364, Aug. 2010.
- [22] W. Zeng, D. Li, and Q. Yin, "Distance and similarity measures between hesitant fuzzy sets and their application in pattern recognition," *Pattern Recognit. Lett.*, vol. 84, pp. 267–271, Dec. 2016.
- [23] D. Caglar, C. Moharram, D. Orhan, and K. Geylani, "Modelling contiki-based IoT systems," in *Proc. 6th Symp. Lang., Appl. Technol.*, vol. 56, 2017.



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